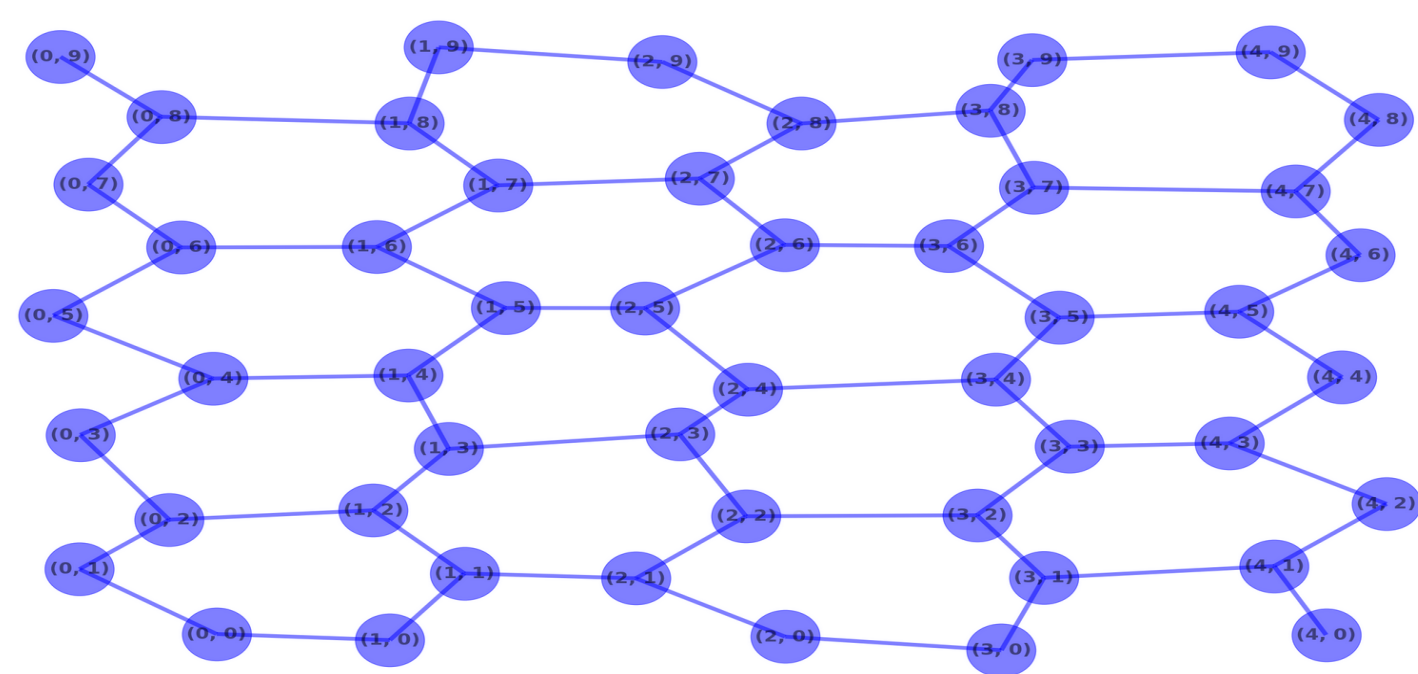
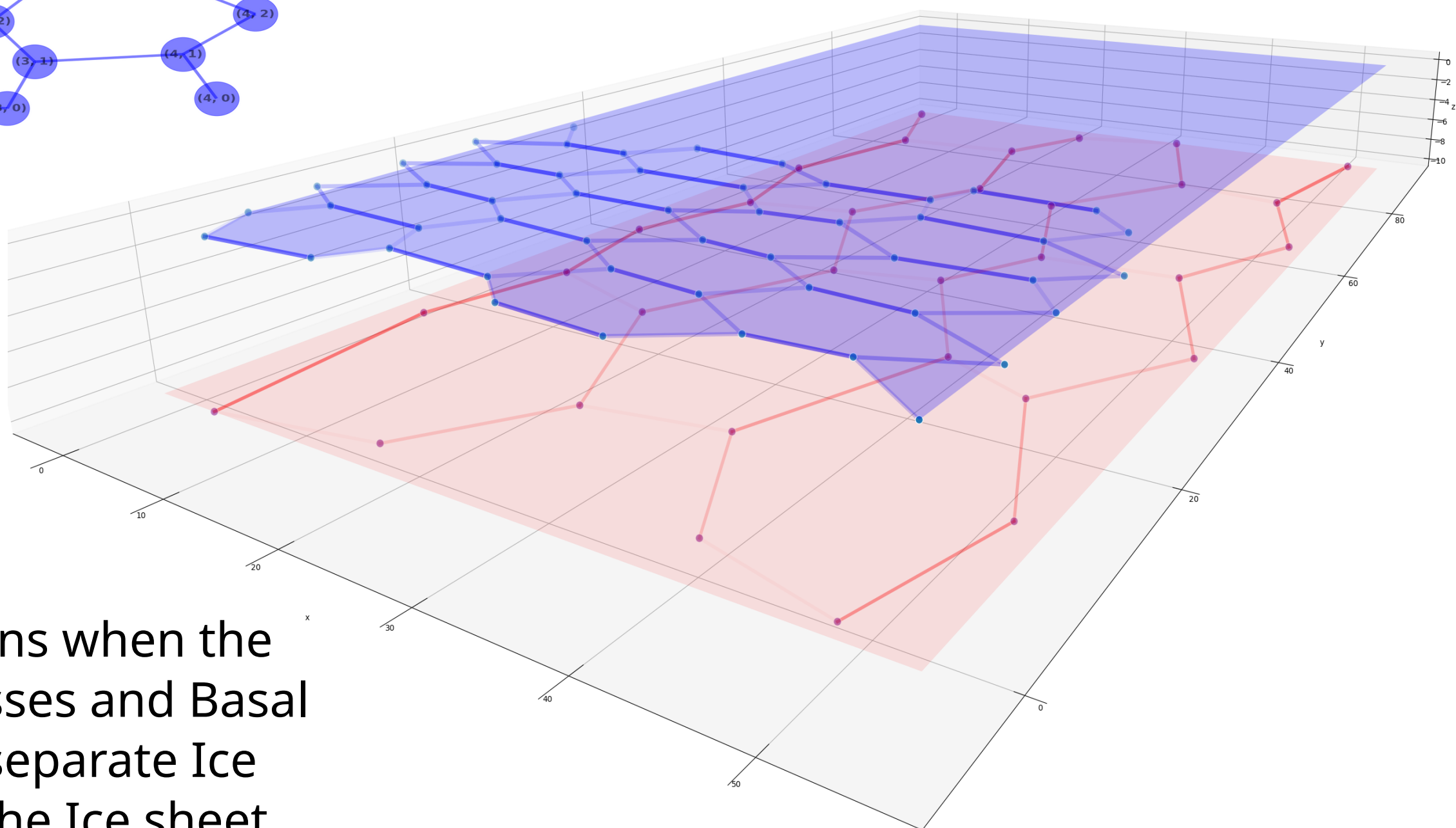


A Quantum Approach to Calving

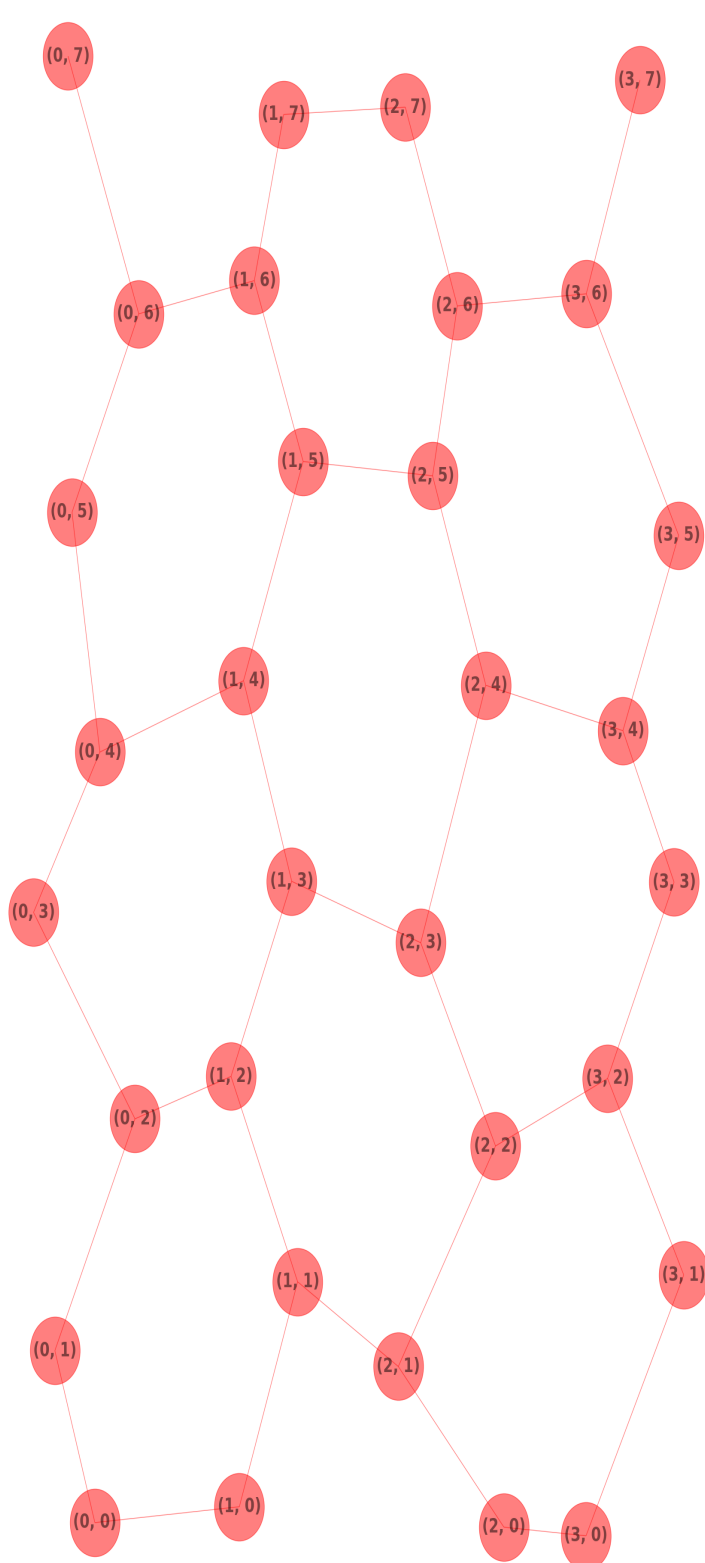
author: C Fillekes (c) 2023 IBM



A Crevasse Network



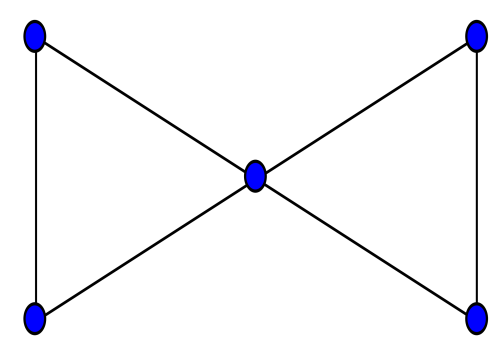
A Network of Basal Cracks and Caverns in the Grounding Zone



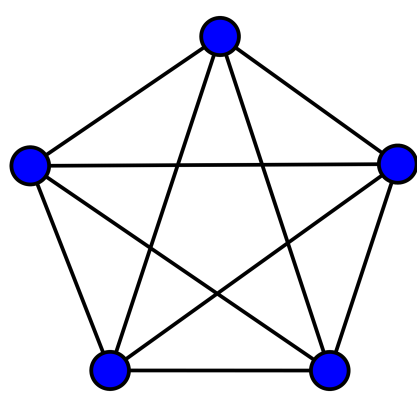
AN HYPOTHESIS:

Calving happens when the surface Crevasses and Basal voids JOIN to separate Ice masses from the Ice sheet...

...**BUT** while the graph of cracks, caverns and crevasses on each surface is necessarily **PLANAR**, the graph connecting these two graphs in the body of the ice is generally **NON-PLANAR**



Planar graphs can be embedded in the plane. Every crossing is a node. Classical flow and cut optimization algorithms apply most effectively to planar graphs.



Non-Planar graphs are where the edges cross each other in any projection onto the plane. Optimizing flows and cuts on large graphs is considered an NP-hard problem.

Derive the weight matrices applied to each edge & node from the Hamiltonian
with some undetermined coefficients to vary in the quantum computing experiments
Use data on crevasses in ice sheets and glaciers to construct the model crack networks



The Quantum Approximate Optimization Algorithm: QAOA

- Map graph node attributes to qubits: 1-1
- Map graph edge attributes to quantum circuits connecting them
- Set up circuits on quantum computer with those mappings (non-optimal where non-planar graphs are mapped to planar arrays of qubits)
- Successive 'shots' through the quantum circuit determine which crack patterns are most energetically likely, including calving
- Suited to modeling episodic calving, numerical stability analysis

